

SHA-1 Hash Function

Practical / Laboratory Program

1. Aim

To implement the SHA-1 hash function and generate a 160-bit message digest for a given input.

2. Algorithm

1. Start the program.
2. Read the input message from the user.
3. Initialize the SHA-1 hash object.
4. Encode the input message into bytes.
5. Update the hash object with the input bytes.
6. Compute the SHA-1 digest.
7. Display the hexadecimal hash value.
8. Stop the program.

3. Program

```
import hashlib

message = input("Enter the message: ")

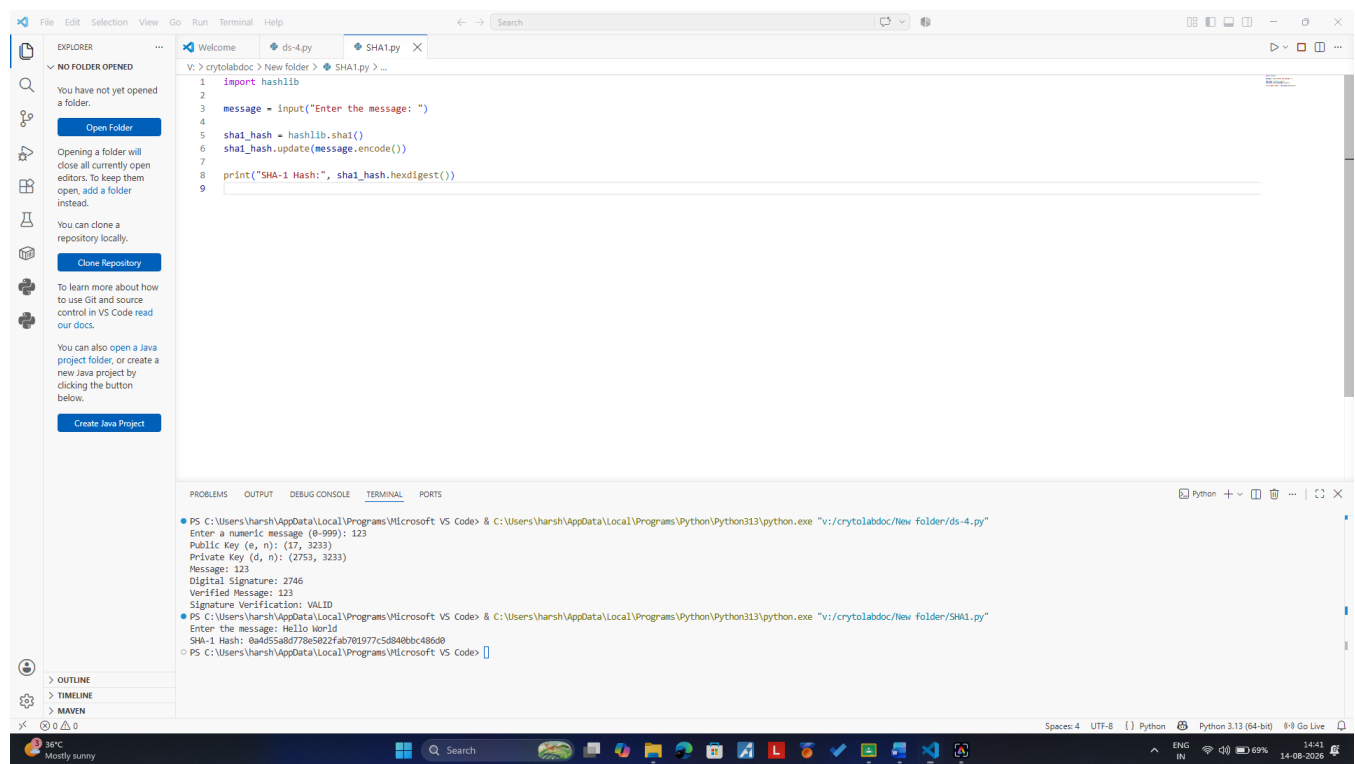
sha1_hash = hashlib.sha1()
sha1_hash.update(message.encode())

print("SHA-1 Hash:", sha1_hash.hexdigest())
```

4. Input

Hello World

5. Output



The screenshot shows a Visual Studio Code editor window with a Python file named `SHA1.py`. The script imports the `hashlib` module, prompts the user for a message, and calculates its SHA-1 hash. The terminal output shows the execution of the script twice: first with a numeric message '123' and then with the text 'Hello World', both resulting in a valid signature verification.

```
1 import hashlib
2
3 message = input("Enter the message: ")
4
5 sha1_hash = hashlib.sha1()
6 sha1_hash.update(message.encode())
7
8 print("SHA-1 Hash:", sha1_hash.hexdigest())
9
```

Terminal Output:

```
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code> & C:\Users\harsh\AppData\Local\Programs\Python\Python313\python.exe "v:/cryptolabdoc/new_folder/ds-4.py"
Enter a numeric message (0-999): 123
Public Key (e, n): (17, 3233)
Private Key (d, n): (2753, 3233)
Message: 123
Digital Signature: 2746
Verified Message: 123
Signature Verification: VALID
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code> & C:\Users\harsh\AppData\Local\Programs\Python\Python313\python.exe "v:/cryptolabdoc/new_folder/SHA1.py"
Enter the message: Hello World
SHA-1 Hash: 0a4d55a8d778e5022fab701977c5d9408bc406d0
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code>
```

6. Result

Thus, the SHA-1 hash function was implemented successfully and the 160-bit hexadecimal message digest was generated.